

I&C Technical Challenge

Company Context

Octavia Carbon relies on precise temperature monitoring across its direct air capture (DAC) systems. While off-the-shelf industrial temperature modules are often used for this purpose, some applications demand custom-designed PCB interfaces for connecting Resistance Temperature Detectors (RTDs) to our control systems.

Scenario

You are tasked with designing a **single-channel RTD input module** to support precise temperature measurement using both **Pt100** and **Pt1000** RTDs in a 4-wire configuration. This design will feed into Octavia Carbon's modular control system based on microcontrollers.

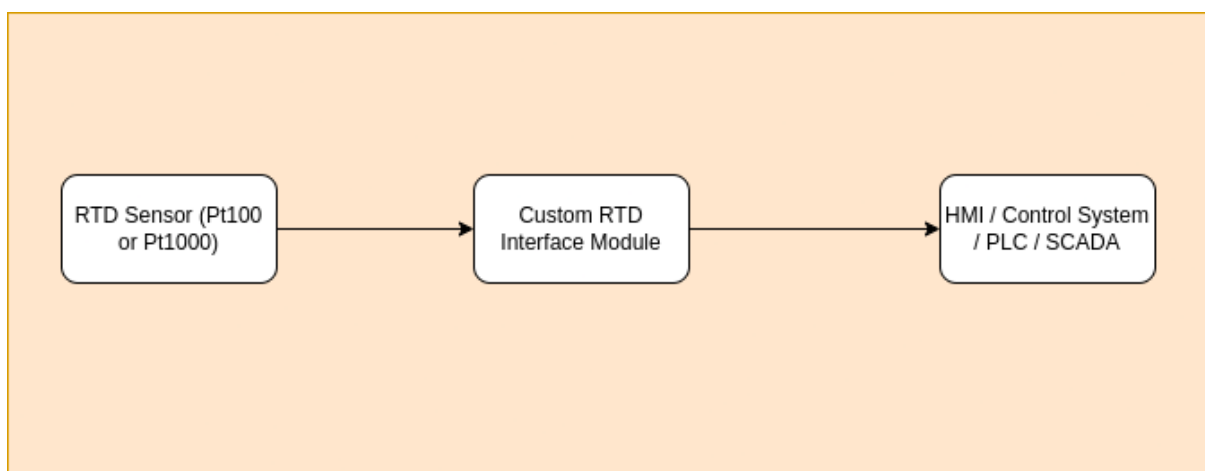
Design Requirements

Your design must:

- Support both **Pt100** and **Pt1000** RTDs (with 3850 ppm/°C alpha).
- Use the **4-wire measurement method**.
- Minimize **self-heating** of the RTD.
- Provide **robust filtering** against electrical noise.
- Deliver a **digital temperature output over SPI**.
- Operate at standard microcontroller logic levels.
- Remain electrically and thermally stable while operating in an environment with ambient temperatures up to 101 °C.

Process Context

Here is a simplified **Process Flow Diagram** for your reference:



The Challenge

Design the **electrical schematic** and implement a **basic embedded driver** for this single-channel RTD input system.

Tasks & Deliverables

1. Provide a clear, complete electrical schematic showing all necessary components and connections and any other required support circuitry (no hand drawings).
2. A description of how your design:
 - a. accommodates **both** Pt100 and Pt1000 sensors.
 - b. How you mitigate noise and minimize self-heating.
 - c. Ensures reliable operation at high ambient temperatures (up to 101 °C), including any thermal design considerations or other protective measures as appropriate.
3. Implementation of a basic firmware component that:
 - a. **Initializes SPI communication** with your selected schematic components.
 - b. **Reads raw measurement data** from the Pt100 and Pt1000 sensor via SPI.
 - c. **Converts the raw data** into a temperature value using the RTD type.
 - d. **(Optional Bonus):** Include a mechanism that selects the appropriate RTD type (Pt100 or Pt1000) and adjusts scaling accordingly.

Evaluation Criteria

- Schematic Completeness
- Design Justification
- Firmware Logic
- Clarity and Documentation

Purpose of the Challenge

In this stage of the interview process, we are evaluating:

- a. Your ability to apply **first principles thinking** to hardware and embedded systems design
- b. Your attention to **electrical engineering best practices**
- c. Your clarity in **communicating technical decisions** and documenting design rationale

We deliberately keep the requirements somewhat open-ended — we are not looking for a single "correct" answer, but rather for your **reasoning process, problem-solving approach, and ability to document and explain your design** clearly.

Submission Instructions

Submit a **single PDF file** that includes:

- Your complete schematic (exported from a schematic capture tool, no hand-drawn schematics).
- Written explanations for your design choices (maximum 1–2 pages).
- Source code in **C** or **Python**, either:
 - Embedded in the PDF, or
 - As links to an accessible GitHub repo (provide link to the same in the PDF).
- (Optional Bonus) Describe any mechanisms for RTD type selection if implemented.

Submission Method:

- Reply directly to the same email thread that sent you this challenge.
- Attach the PDF to your reply.
- Do not start a new email.
- Do not start a new thread.

This ensures that your submission is correctly linked to your application.

Submission Deadline: Sunday, 4th May 2025